Analysis of Thermal Budget and Energy Efficiency: Standard High-Temperature vs. Ultraviolet-Assisted Chemical Vapor Deposition for Silicon Carbide Epitaxy

Project Report

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CERTIFICATE OF COMPLETION

This is to certify that the work contained in this project titled, **Analysis of Thermal Budget and Energy Efficiency: Standard High-Temperature vs. Ultraviolet-Assisted CVD for Silicon Carbide Epitaxy** submitted by Ruchir Saraf, with a college ID I22PH048, to the Department of Physics, Sardar Vallabhbhai National Institute of Technology, Surat, is a bonafide record of the work done. The project satisfies the requirements of the rules and regulations related to the fourth year mini project III of the integrated 5-year Master of Science (Physics) for the academic year 2025-26 and is considered satisfactory.

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Examiner 1

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Examiner 2

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Abstract

This mini project presents a detailed thermal and energetic analysis of silicon carbide (SiC) epitaxy using conventional high-temperature chemical vapor deposition (HTCVD) and a proposed ultraviolet-assisted CVD (UV-CVD) approach. A steady-state thermal model of a vertical cold-wall reactor is developed to quantify radiative and convective losses and to evaluate the impact of shifting the activation mechanism from purely thermal to photonic. For a 4-inch SiC-coated graphite susceptor, results show that radiative losses dominate the thermal budget at 1600°C, scaling with the fourth power of temperature and leading to a total power demand of about 7 kW per reactor. By introducing UV-driven photolysis of silane and propane, the required substrate temperature can be reduced to around 900 K, lowering radiative losses by approximately 94.7% and the overall process power to about 1.6 kW, even after accounting for realistic UV source inefficiencies. Beyond energy savings of roughly 76.5%, the lower-temperature regime is expected to mitigate thermal stress, reduce slip-line defects, and suppress dopant memory effects arising from reactor wall outgassing. The study concludes that UV-CVD offers a physically grounded pathway toward energy-efficient, higher-yield SiC epitaxy and has strong potential as a key enabling technology for next-generation wide bandgap power electronics and green semiconductor manufacturing.

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Chapter 1

Introduction

The contemporary semiconductor landscape is undergoing fundamental restructuring, driven by the imperative for electrification and energy efficiency in power electronics. Silicon Carbide (SiC), a wide bandgap material, exhibits vastly superior properties compared to silicon for high-voltage, high-temperature, and high-frequency applications . However, the manufacturing complexity of SiC epitaxial growth remains a significant bottleneck, with the incumbent High-Temperature Chemical Vapor Deposition (HTCVD) technology imposing severe energy penalties and inducing crystallographic defects.

This research presents an exhaustive theoretical and numerical analysis comparing traditional HTCVD against a novel Ultraviolet-Assisted Chemical Vapor Deposition (UV-CVD) methodology. The primary objective is to quantify the thermal budget implications of shifting from thermal energy (phonons) to photonic energy (photons) as the primary activation mechanism.

By developing a steady-state thermal model of a vertical cold-wall reactor, this study evaluates the feasibility of reducing substrate temperatures from the standard $1600°C$ (1873) to a photon-enhanced regime of $600°C - 900°C$ (873).

Key Finding: The analysis indicates that radiative heat loss—which scales with the fourth power of temperature (T^4)—dominates energy consumption at high temperatures. By leveraging UV photons to dissociate precursor molecules such as Silane (SiH_4) and Propane (C_3H_8), substrate temperature can be reduced to 900,

resulting in a **94.7% reduction in radiative heat loss** . Even accounting for UV source inefficiencies (20% wall-plug efficiency), net process energy consumption is reduced by approximately **43% to 76.5%** depending on reactor configuration.

Beyond thermodynamic savings, this transition mitigates thermal stress-induced defects (slip lines) and suppresses the "memory effect" caused by dopant outgassing from reactor walls, yielding significant improvements in device yield.

Chapter 2

2. The Strategic Imperative of Wide Bandgap Epitaxy

2.1 The Limitations of Silicon in the Power Era

For over five decades, silicon has been the substrate of choice for microelectronics. However, silicon's physical properties impose hard limits on power applications. The bandgap of silicon (1.12) results in a relatively low critical breakdown field (≈ 0.3) . In high-voltage applications such as electric vehicle traction inverters or grid-scale solar inverters, thick, highly resistive drift layers are necessary to block voltage, increasing On-State Resistance ($R_{DS(on)}$) and conduction losses.

Silicon Carbide (4H-SiC polytype) offers a bandgap of 3.26 and a critical breakdown field nearly ten times that of silicon (≈ 2.5) . This enables fabrication of unipolar devices (MOSFETs, Schottky Diodes) that are thinner, faster, and capable of higher-temperature operation. The global SiC market is experiencing robust growth driven by increasing demand for SiC-based power electronics in electric vehicles, renewable energy infrastructure, and high-power applications.

However, unlike silicon—which can be grown from a melt into massive, dislocation-free boules via the Czochralski method—SiC does not melt at atmospheric pressure; it sublimes at approximately $2830°C$. This necessitates growth of bulk substrates via Physical Vapor Transport (PVT), yielding wafers with high defect densities (micropipes, basal plane dislocations). Consequently, active device layers are grown on top of the substrate via epitaxy, a process requiring precise control.

2.2 The High-Temperature Conundrum (HTCVD)

The industry standard for depositing epitaxial layers is Chemical Vapor Deposition (CVD). The process involves transporting precursor gases—typically Silane (SiH_4) for silicon and Propane (C_3H_8) for carbon—over a heated substrate.

The fundamental challenge lies in the robust nature of Si-C and C-H bonds. To achieve high-quality step-flow growth, adatoms must have high surface mobility. Furthermore, decomposition of propane and silane into active radical species requires significant activation energy. In standard thermal CVD, this energy is provided by heating the graphite susceptor to temperatures between $1500°C$ and $1650°C$.

This high-temperature regime creates cascading engineering challenges:

- **Thermodynamic Inefficiency:** Maintaining a susceptor at $1600^{\circ}C$ in a cold-wall reactor is energetically inefficient due to radiative losses scaling with T^4.
- **Mechanical Stress:** Radial thermal gradients induce significant shear stresses. When these exceed the Critical Resolved Shear Stress (CRSS) of SiC, plastic deformation occurs, manifesting as slip lines and wafer bowing.
- **Impurity Memory:** At $1600^{\circ}C$, dopants from previous runs can desorb from chamber walls and incorporate into growing films, complicating doping transitions.

2.3 The Photonic Solution: UV-CVD

Ultraviolet-Assisted CVD represents a paradigm shift. Instead of relying on vibrational energy of heat to break chemical bonds (pyrolysis), UV-CVD utilizes electronic energy of photons (photolysis). If the photon energy ($h\nu$) matches or exceeds the bond dissociation energy of precursor molecules, bonds can be severed directly, regardless of substrate temperature..

This decoupling of "chemistry" from "temperature" allows the substrate to be heated only to the point required for surface diffusion and crystal ordering ($600^{\circ}C - 900^{\circ}C$), rather than the extreme temperatures required for gas-phase cracking . This report validates this hypothesis through rigorous thermal and quantum mechanical analysis.

Chapter 3

3. Theoretical Framework: Thermodynamics and Heat Transfer

To accurately quantify the thermal budget, we establish the physical laws governing heat dissipation in a CVD reactor. The system is modeled as a Vertical Cold-Wall Reactor, the standard configuration for research and high-value production.

3.1 Radiative Heat Transfer Physics

In vacuum or low-pressure environments, radiation is the dominant heat transfer mechanism at high temperatures. For a blackbody, total emissive power is given by the Stefan-Boltzmann law:

$$E_b = \sigma T^4$$

Real materials are "gray bodies" characterized by emissivity (ϵ), the ratio of radiation emitted to radiation emitted by a blackbody at the same temperature. Net radiative heat exchange between a hot susceptor and cold reactor walls is described by:

$$P_{rad} = \frac{\sigma A_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1-\epsilon_2}{\epsilon_2}(\frac{A_1}{A_2})}$$

Given that reactor walls (A_2) are significantly larger than the susceptor (A_1), the ratio A_1/A_2 approaches zero, and the equation simplifies to:

$$P_{rad} = \epsilon_{sus}\sigma A_{sus}(T_{sus}^4 - T_{wall}^4)$$

Where:

- P_{rad} = net radiative power loss (Watts)
- σ = Stefan-Boltzmann constant ($5.67 \times 10^{-8} W/m^2K^4$)
- ϵ_{sus} = emissivity of SiC-coated graphite susceptor
- A_{sus} = surface area of susceptor (m^2)
- T_{sus} and T_{wall} = absolute temperatures (Kelvin)

Theoretical Insight: The T^4 dependence is critical. Doubling temperature from 800 to 1600 increases radiative loss by a factor of $2^4 = 16$. This non-linearity is the root cause of HTCVD inefficiency.

3.2 Convective and Conductive Heat Transfer

CVD processes typically operate at reduced pressures ($50 - 100$) to suppress gas-phase nucleation and improve uniformity. In this transition regime, heat transfer combines conduction through gas boundary layer and forced convection from gas flow.

For steady-state energy balancing, this is approximated using Newton's Law of Cooling:

$$P_{gas} = h_{conv}A_{sus}(T_{sus} - T_{gas})$$

In hydrogen-rich environments at low pressure, h_{conv} is estimated at approximately 100 . Unlike radiation, convective loss scales linearly with temperature difference (ΔT), making it

negligible at high temperatures but more significant at lower temperatures.

3.3 The Thermal Budget Equation

Total power required to maintain steady-state growth temperature is the sum of all losses, adjusted for heating system efficiency:

$$P_{thermal} = \frac{P_{rad} + P_{gas}}{\eta_{heater}}$$

Assuming inductive heating efficiency (η_{heater}) of 85%, this accounts for losses in RF generator, copper coils, and coupling efficiency to the graphite susceptor.

Chapter 4

4. Chemical Kinetics: Pyrolysis vs. Photolysis

Understanding the energy balance requires understanding why high temperatures are used in the first place, and how UV photons circumvent this requirement.

4.1 Arrhenius Kinetics in Thermal CVD

In standard HTCVD, reaction rate ($\dot{k}$) is governed by the Arrhenius equation:

$$\dot{k} = A\exp\left(\frac{-E_a}{k_B T}\right)$$

For SiC growth using Silane and Propane, the reaction involves multiple steps: gas-phase decomposition of precursors, transport to surface, adsorption, surface diffusion to step edges, and desorption of byproducts . The limiting step usually has high activation energy (E_a) ranging from 20 to 50 kcal/mol. The exponential dependence means that below a critical temperature (roughly $1400°C$ for SiC), reaction rate drops to near zero.

4.2 Quantum Mechanics of UV-Assisted Epitaxy

UV-CVD replaces thermal activation ($k_B T$) with direct photon activation ($h\nu$). When a photon with sufficient energy strikes a molecule, it excites an electron from a bonding orbital to an anti-bonding orbital, causing instantaneous dissociation.

Photon Energy Calculation:

The energy of a photon is inversely proportional to its wavelength:

$$E = \frac{hc}{\lambda}$$

For an Argon Fluoride (ArF) Excimer source emitting at $\lambda = 193$:

$$E_{193nm} \approx 6.4$$

Bond Dissociation Analysis:

Comparing photon energy to bond strengths of precursors :

- Si-H bond (in Silane): ~ 3.2
- C-H bond (in Propane): ~ 4.3
- Si-C bond (Lattice): $\sim 3.0 - 4.5$

Since 6.4, a single 193 nm photon carries sufficient energy to break any of these bonds . This makes precursor decomposition temperature-independent. The reaction becomes "photon limited" rather than "thermally activated," decoupling chemistry from temperature requirements .

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5. Simulation Methodology and Reactor Architecture

5.1 Physical Model Parameters

The simulation assumes a single-wafer Vertical Cold-Wall CVD reactor—the most common configuration for high-precision epitaxy. This geometry minimizes the "memory effect" and allows for uniform gas flow .

Parameter	Value	Description
Susceptor Radius (r)	0.05 (50 mm)	Supports standard 4-inch (100 mm) wafer
Susceptor Area (A)	0.00785	Derived from πr^2
Susceptor Material	SiC-Coated Graphite	High purity, high thermal conductivity
Emissivity (ϵ)	0.85	Average value for graphite/SiC at high T
Wall Temperature (T_{wall})	300 ($27^{\circ}C$)	Water-cooled stainless steel
Convective Coefficient (h)	100	Estimate for low-pressure H_2 flow
Heater Efficiency (η_{heater})	0.85 (85%)	Inductive coupling efficiency

Table 1: Reactor Geometric and Material Parameters

5.2 Simulation Scenarios

The study compares two distinct operational modes:

Scenario A: Standard HTCVD

- Target Temperature: $1600^{\circ}C$ (1873)
- Mechanism: Thermal decomposition

- Energy Inputs: RF Heating only

Scenario B: UV-Assisted CVD

- Target Temperature: $627°C$ (900)
- Mechanism: Photolytic decomposition + Thermal surface diffusion
- Energy Inputs: RF Heating (reduced) + UV Source

5.3 UV Source Modeling Assumptions

For Scenario B, the energy cost of the UV source must be accounted for :

- **Required Intensity:** 1 at wafer surface
- **Source Efficiency:** 20% wall-plug efficiency (Excimer lamps or lasers)
- **Optical Losses:** 50% of generated light lost in optical train before reaching wafer

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6. Numerical Analysis and Results

6.1 Scenario A: High-Temperature Growth (1873 K)

Step 1: Radiative Loss Calculation

$$P_{rad} = 0.85 \times (5.67 \times 10^{-8}) \times 0.00785 \times (1873^4 - 300^4)$$

Analyzing temperature terms :

- $1873^4 \approx 1.230 \times 10^{13}$
- $300^4 \approx 0.008 \times 10^{13}$
- $(T_{sus}^4 - T_{wall}^4) \approx 1.229 \times 10^{13}$

The wall temperature contribution is negligible; radiation is entirely dominated by susceptor temperature .

$$P_{rad} \approx 3.78 \times 10^{-10} \times 1.229 \times 10^{13} \approx 4{,}650$$

Step 2: Convective Loss Calculation

$$P_{gas} = 100 \times 0.00785 \times (1873 - 300) = 0.785 \times 1573 \approx 1{,}235$$

Step 3: Total Power Input (Scenario A)

Net thermal output required:

$$P_{out} = 4650 + 1235 = 5{,}885$$

With 85% efficiency:

$$P_{input_A} = \frac{5885}{0.85} \approx 6{,}923$$

Analysis: The standard process consumes nearly 7 kW for a single 4-inch wafer run. 67% of thermal energy is lost purely to radiation .

6.2 Scenario B: UV-Assisted Low-Temperature Growth (900 K)

Step 1: Radiative Loss Calculation

With temperature reduced to 900 ($627°C$) :

$$P_{rad} = 0.85 \times (5.67 \times 10^{-8}) \times 0.00785 \times (900^4 - 300^4)$$

Temperature terms :

- $900^4 \approx 6.56 \times 10^{11}$
- $(900^4 - 300^4) \approx 6.48 \times 10^{11}$

$$P_{rad} \approx 3.78 \times 10^{-10} \times 6.48 \times 10^{11} \approx 245$$

Finding: Radiative loss drops from 4,650 W to 245 W—a **94.7% reduction** .

Step 2: Convective Loss Calculation

$$P_{gas} = 100 \times 0.00785 \times (900 - 300) = 0.785 \times 600 = 471$$

Step 3: UV Source Power Requirement

Target optical power on wafer: 1

Total electrical power for lamp :

$$P_{lamp_electric} = \frac{P_{optical}}{\eta_{lamp} \times \eta_{optics}} = \frac{78.5}{0.20 \times 0.50} = \frac{78.5}{0.1} = 785$$

Step 4: Total Power Input (Scenario B)

Thermal heater input:

$$P_{thermal_out} = 245 + 471 = 716$$

$$P_{thermal_input} = \frac{716}{0.85} \approx 842$$

Total system power :

$$P_{total_B} = 842 + 785 = 1{,}627$$

Analysis: The UV-assisted process consumes roughly 1.6 kW. Even with the inefficient UV source, total energy footprint is drastically lower .

6.3 Comparative Energy Analysis

Energy Component	**Standard CVD (1873 K)**	**UV-CVD (900 K)**	**Difference**	**% Impact**
Radiative Loss	4,650 W	245 W	−4,405 W	−94.7%
Convective Loss	1,235 W	471 W	−764 W	−61.8%
Heater Input Power	6,923 W	842 W	−6,081 W	−87.8%
UV Source Power	0 W	785 W	+785 W	N/A
TOTAL POWER	**6,923 W**	**1,627 W**	−5,296 **W**	−76.5%

Table 2: Comparative Thermal Budget Analysis

The data confirms that the energy penalty of the UV source (785 W) is negligible compared to massive savings in radiative heat

loss (4,405 W). The UV-CVD process operates at approximately **23.5% of the energy cost** of the standard process .

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7. Second-Order Effects: Yield and Quality Engineering

While energy savings are substantial, the primary driver for adopting UV-CVD in high-value semiconductor manufacturing may be yield enhancement. The reduction in process temperature has profound implications for crystalline quality .

7.1 Mitigation of Thermal Stress and Slip Lines

Silicon Carbide is extremely hard, but at $1600°C$, it becomes susceptible to plastic deformation. In standard CVD reactors, wafer temperature is rarely perfectly uniform :

- **Radial Gradients:** Wafer edges often cool faster than center due to radiation view factors
- **Vertical Gradients:** Sharp gradient between hot wafer surface and cooler gas inlet

These gradients create internal stress fields ($\sigma_{thermal}$). If stress exceeds the Critical Resolved Shear Stress (CRSS), the crystal lattice "slips" to relieve tension, creating arrays of dislocations known as slip lines .

Impact: Slip lines physically deform the wafer (bowing/warping), making it impossible to process in subsequent lithography steps (requiring nanometer-level flatness). They also act as killing defects for devices .

UV-CVD Advantage: Operating at $627°C$ (900 K), the material remains deep within its elastic regime. The CRSS at 900 K is

significantly higher than at 1873 K. Furthermore, the absolute magnitude of radiative cooling is lower, naturally reducing radial temperature gradients. This theoretical reduction in stress suggests UV-CVD could increase usable area of large-diameter (150 mm/200 mm) wafers by 5-10% .

7.2 Suppression of the "Memory Effect"

Doping control is critical for power devices. A typical SiC MOSFET requires a drift layer with low nitrogen concentration ($10^{15} cm^{-3}$) grown on top of a substrate with high concentration ($10^{18} cm^{-3}$) .

The Problem: At $1600°C$, graphite insulation and reactor walls outgas impurities absorbed during previous runs, creating a background doping "fog" that prevents sharp transitions .

UV-CVD Advantage: With the susceptor at only $627°C$ and minimal radiative heating (245 W vs 4,650 W), reactor walls remain cold. Desorption of contaminants is exponentially suppressed, allowing for "steep" doping profiles and enabling more complex, efficient device architectures .

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8. Engineering Challenges and Industrial Outlook

Despite overwhelming thermodynamic arguments, UV-CVD faces practical hurdles to mass adoption .

8.1 The "Window Clouding" Phenomenon

For UV light to enter the reactor, a transparent window (Quartz or Sapphire) is required. However, process gases (SiH_4) are present inside the chamber .

Issue: If the window surface interacts with process gas, a thin film of Silicon or SiC will deposit on the window. Even a few nanometers of deposition renders the window opaque to 193 nm light, effectively cutting off the power source .

Solution: Advanced "Purge Gas Curtains" are required, designing reactor flow dynamics such that a high-velocity stream of inert gas (Argon) constantly sweeps the inner surface of the window, creating a barrier preventing reactant gases from contacting the optics .

8.2 Growth Rate and Throughput

Standard thermal CVD achieves growth rates of $10 - 20$. UV-CVD rates are currently limited by photon flux, typically achieving $0.5 - 3$. While power per second is lower for UV-CVD, time per wafer is longer.

Trade-off: If the process takes 10 times longer, total energy savings per wafer might diminish .

Hybrid Future: The likely industrial solution is a hybrid approach. UV-CVD may be used for critical, thin active layers where quality and doping sharpness are paramount, while traditional thermal CVD is used for thick, bulk drift layers where speed is the priority .

8.3 Economic Context: Total Cost of Ownership

Energy savings of ~ 15 kWh per run translate to roughly $2.00 - $3.00 per wafer in direct electricity costs. While seemingly small, for a fab running 100,000 wafers per year, this is substantial . However, true economic value lies in Capital Expenditure (CapEx) reduction :

- **Cooling Infrastructure:** A fab designed for UV-CVD requires 75% less cooling capacity—smaller water chillers, HVAC units, and reduced power distribution infrastructure
- **Green Manufacturing:** As the semiconductor industry faces pressure to decarbonize, technologies offering 75% reduction in Scope 2 emissions (electricity usage) will become highly attractive for corporate sustainability goals

9. Conclusion

This research report successfully expands upon initial theoretical models to provide a comprehensive evaluation of UV-Assisted Epitaxy for Silicon Carbide.

Key Conclusions

1. **Thermodynamics:** Radiative heat loss is the dominant energy sink in standard CVD ($1600°C$), scaling aggressively with T^4 .
2. **Energy Efficiency:** Reducing process temperature to 900 via UV assistance reduces total thermal budget by **76.5%**, from $\sim$ 7 kW to $\sim$ 1.6 kW per reactor, even accounting for UV source inefficiency .
3. **Yield Enhancement:** The low-temperature regime offers significant secondary benefits, including reduction of slip-line defects and elimination of doping memory effects .
4. **Industrial Viability:** UV-CVD is not merely an academic curiosity but a viable pathway toward "Green Manufacturing." It addresses fundamental thermodynamic inefficiencies of high-temperature processing while offering solutions to thermal stress limitations plaguing the transition to larger wafer diameters .

As UV source technology matures and flux densities increase, UV-CVD is poised to become a cornerstone technology for the next generation of wide bandgap power electronics. The combination of energy efficiency, improved crystal quality, and reduced manufacturing infrastructure represents a transformative advancement in semiconductor processing .

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